

CO1-AT1-SHORT ANSWER TEST

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COURSE CODE: ITA 0502

COURSE NAME: COMPUTER VISION.

1) low spatial resolution means the image is sampled with fewer pixels, so small objects become unclear. increasing the sampling rate/resolution by using more pixels improves the image details.

2) we use image enhancement and noise filtering such as histogram equalization and a median filter. These techniques increase brightness/contrast and reduce noise.

3) Quantization converts continuous intensity values into a limited number of digital intensity levels. it helps represent the image digitally so that it can be stored and processed by a computer.

4) we use contrast enhancement such as histogram equalization. it improves the contrast between the product and defects, making defects easier to identify.

5) image segmentation divides the image into regions such as roads, vehicles, and pedestrians. This helps the self-driving car understand its

surroundings and make safe driving decisions

- 6) Edge detection identifies sharp changes in intensity that represent object boundaries. It can highlight cracks or edges, helping the inspection system detect defects.
- 7) A median filter can be used to reduce unwanted noise while preserving important edges. This makes the tumor region clearer for analysis.
- 8) Object detection identifies vehicles in the image and determines their positions. Tracking these vehicles helps monitor traffic flow, congestion, and road conditions.
- 9) The blur is caused by object movement during image acquisition. Motion deblurring techniques can be applied to restore the sharpness of the image.
- 10) Image classification assigns products to classification to categorize product as defective or good based on their visual features. This supports automatic quality inspection and reduces manual checking.